

ECE 786 Final Project

Profile-Based Cache Bypassing in Accel-Sim

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1. Cache Efficiency Analysis

Six benchmarks were evaluated using the ECE786_NCSU configuration (based on SM2_GTX480) with a 1M instruction cap. Each benchmark was run twice: once with L1D bypassing disabled (-gpgpu_gmem_skip_L1D 0) and once fully enabled (-gpgpu_gmem_skip_L1D 1). Benchmarks are classified as Cache Unfriendly (IPC improves $\geq 4\%$ with bypass), Cache Friendly (IPC degrades $\geq 4\%$), or Cache Insensitive (IPC changes less than 4%).

BFS exhibits mixed behavior — early kernels (uid 1-4) are cache-friendly as they process densely connected graph nodes with good spatial locality, while later kernels (uid 5-8) become cache-unfriendly processing sparse frontier nodes. LPS shows the most pronounced cache-unfriendly behavior at +46.8%. LUD's lud_internal kernels are consistently cache-unfriendly in early iterations, becoming insensitive in later iterations as the matrix subproblem shrinks.

Table 1: Cache Efficiency Analysis Results

Benchmark	Kernel Name	UID	IPC	IPC w/ Bypass	Change %	Category
LPS	_Z13GPU_laplace3diiiiiPfs_	1	371.5598	545.6426	+46.9%	Unfriendly
BFS	_Z6KernelP4NodePiPbS2_S1_S2_i	1	79.5269	67.3758	-15.3%	Friendly
BFS	_Z6KernelP4NodePiPbS2_S1_S2_i	2	72.5531	59.6617	-17.8%	Friendly
BFS	_Z6KernelP4NodePiPbS2_S1_S2_i	3	65.0414	50.9865	-21.6%	Friendly
BFS	_Z6KernelP4NodePiPbS2_S1_S2_i	4	54.4065	40.2290	-26.1%	Friendly
BFS	_Z6KernelP4NodePiPbS2_S1_S2_i	5	25.3154	32.2584	+27.4%	Unfriendly
BFS	_Z6KernelP4NodePiPbS2_S1_S2_i	6	17.6819	31.5100	+78.2%	Unfriendly
BFS	_Z6KernelP4NodePiPbS2_S1_S2_i	7	19.1406	40.9166	+113.8%	Unfriendly
BFS	_Z6KernelP4NodePiPbS2_S1_S2_i	8	34.4767	56.8447	+64.9%	Unfriendly
BFS	_Z6KernelP4NodePiPbS2_S1_S2_i	9	115.1721	113.7748	-1.2%	Insensitive
NQU	_Z24solve_nqueen_cuda_kerneliiiPjS_S_S_i	1	34.2939	34.3879	+0.3%	Insensitive
BP	_Z22bpnn_layerforward_CUDAPfs_S_S_ii	1	665.0260	712.9353	+7.2%	Unfriendly
BP	_Z24bpnn_adjust_weights_cudaPfiS_iS_S_	2	129.7257	126.9964	-2.1%	Insensitive
HS	_Z14calculate_tempiPfs_S_iiiiiiiiii	1	356.3531	366.1858	+2.8%	Insensitive
LUD	_Z12lud_diagonalPfii	1	0.8307	0.8342	+0.4%	Insensitive
LUD	_Z13lud_perimeterPfii	2	21.9486	22.8194	+4.0%	Unfriendly
LUD	_Z12lud_internalPfii	3	279.1061	326.3870	+16.9%	Unfriendly
LUD	_Z12lud_diagonalPfii	4	0.8356	0.8389	+0.4%	Insensitive
LUD	_Z13lud_perimeterPfii	5	21.0558	22.0443	+4.7%	Unfriendly
LUD	_Z12lud_internalPfii	6	261.4349	317.9300	+21.6%	Unfriendly
LUD	_Z12lud_diagonalPfii	7	0.8356	0.8389	+0.4%	Insensitive
LUD	_Z13lud_perimeterPfii	8	19.4577	20.5372	+5.5%	Unfriendly
LUD	_Z12lud_internalPfii	9	229.8357	291.5305	+26.8%	Unfriendly
LUD	_Z12lud_diagonalPfii	10	0.8356	0.8389	+0.4%	Insensitive

Benchmark	Kernel Name	UID	IPC	IPC w/ Bypass	Change %	Category
LUD	_Z13lud_perimeterPfii	11	17.9376	18.9236	+5.5%	Unfriendly
LUD	_Z12lud_internalPfii	12	209.5789	255.2592	+21.8%	Unfriendly
LUD	_Z12lud_diagonalPfii	13	0.8356	0.8389	+0.4%	Insensitive
LUD	_Z13lud_perimeterPfii	14	16.5699	17.4015	+5.0%	Unfriendly
LUD	_Z12lud_internalPfii	15	193.1855	230.7529	+19.4%	Unfriendly
LUD	_Z12lud_diagonalPfii	16	0.8356	0.8389	+0.4%	Insensitive
LUD	_Z13lud_perimeterPfii	17	15.2230	15.8610	+4.2%	Unfriendly
LUD	_Z12lud_internalPfii	18	181.3300	206.9248	+14.1%	Unfriendly
LUD	_Z12lud_diagonalPfii	19	0.8356	0.8389	+0.4%	Insensitive
LUD	_Z13lud_perimeterPfii	20	13.7626	14.2946	+3.9%	Insensitive
LUD	_Z12lud_internalPfii	21	154.6429	179.6369	+16.2%	Unfriendly
LUD	_Z12lud_diagonalPfii	22	0.8356	0.8389	+0.4%	Insensitive
LUD	_Z13lud_perimeterPfii	23	12.2285	12.7309	+4.1%	Unfriendly
LUD	_Z12lud_internalPfii	24	131.8909	152.3141	+15.5%	Unfriendly
LUD	_Z12lud_diagonalPfii	25	0.8356	0.8389	+0.4%	Insensitive
LUD	_Z13lud_perimeterPfii	26	10.8091	11.1566	+3.2%	Insensitive
LUD	_Z12lud_internalPfii	27	114.0209	121.8041	+6.8%	Unfriendly
LUD	_Z12lud_diagonalPfii	28	0.8356	0.8389	+0.4%	Insensitive
LUD	_Z13lud_perimeterPfii	29	9.2525	9.5734	+3.5%	Insensitive
LUD	_Z12lud_internalPfii	30	86.4081	93.7268	+8.5%	Unfriendly
LUD	_Z12lud_diagonalPfii	31	0.8356	0.8389	+0.4%	Insensitive
LUD	_Z13lud_perimeterPfii	32	7.7562	7.9756	+2.8%	Insensitive
LUD	_Z12lud_internalPfii	33	65.2366	68.4110	+4.9%	Unfriendly
LUD	_Z12lud_diagonalPfii	34	0.8356	0.8389	+0.4%	Insensitive
LUD	_Z13lud_perimeterPfii	35	6.2395	6.3822	+2.3%	Insensitive
LUD	_Z12lud_internalPfii	36	42.3833	44.4568	+4.9%	Unfriendly
LUD	_Z12lud_diagonalPfii	37	0.8356	0.8389	+0.4%	Insensitive
LUD	_Z13lud_perimeterPfii	38	4.7077	4.7927	+1.8%	Insensitive
LUD	_Z12lud_internalPfii	39	25.6468	26.5146	+3.4%	Insensitive
LUD	_Z12lud_diagonalPfii	40	0.8356	0.8389	+0.4%	Insensitive
LUD	_Z13lud_perimeterPfii	41	3.1367	3.2004	+2.0%	Insensitive
LUD	_Z12lud_internalPfii	42	11.5951	11.8822	+2.5%	Insensitive
LUD	_Z12lud_diagonalPfii	43	0.8356	0.8389	+0.4%	Insensitive
LUD	_Z13lud_perimeterPfii	44	1.5807	1.6007	+1.3%	Insensitive
LUD	_Z12lud_internalPfii	45	2.9456	2.9871	+1.4%	Insensitive
LUD	_Z12lud_diagonalPfii	46	0.8356	0.8389	+0.4%	Insensitive

2. Profile-Based Cache Bypassing

2.1 Implementation

The profile-based bypass was implemented in Accel-Sim by modifying four source files under the `gpgpu-sim/src/` directory.

In `abstract_hardware_model.h`, two boolean flags were added to the kernel configuration struct: `profile_cache` (enables profiling of L1D reference counts and file dump) and `load_profile` (loads profile data and applies bypass logic).

In `gpu-sim.cc`, both flags were registered as command-line options. A call to `dump_profile_all_cores()` was added in `set_kernel_done()` to write per-SM profile files at kernel completion. A call to `load_profile_all_cores()` was added in `gpgpu_sim::init()` to load and merge all SM profile files before simulation begins.

In `shader.h`, the `ldst_unit` class was extended with: `dump_profile()`, `load_profile_data()`, `load_global_profile()` methods; `m_profile_bypass_count` counter; and `m_profile_counters` and `m_global_profile_counters` maps keyed by `(kernel_uid, block_addr)` pairs. Cluster-level wrapper methods `dump_profile_all_cores()`, `load_profile_all_cores()`, and `print_bypass_stats()` were added to `simt_core_cluster`.

In `shader.cc`, two blocks were added at the top of `ldst_unit::memory_cycle()` before the `bypassL1D` declaration. The profiling block records cache block reference counts (`addr >> 7`) for global load instructions when `profile_cache` is enabled. The bypass block, when `load_profile` is enabled, looks up each global load address in the merged global profile map `m_global_profile_counters` and sets `inst.cache_op = CACHE_GLOBAL` if the total reference count across all SMs is less than 3 or if the block was never seen during profiling. Setting `inst.cache_op = CACHE_GLOBAL` ensures the response fill path also correctly bypasses L1D, avoiding crashes in the `SECTOR_ASSOC` cache fill logic. Profile data is saved per-SM to `profile_smN.txt` files. During the bypass run, all 15 SM files are loaded and their counts are merged into a single global map, enabling cross-SM reuse tracking.

2.2 How to Reproduce

Step 1 — Profile Run: Set the following options in `gpgpusim.config`, then run the benchmark from a dedicated output directory. The simulator will generate 15 per-SM profile files (`profile_sm0.txt` through `profile_sm14.txt`) in the current working directory at kernel completion.

```
-gpgpu_gmem_skip_L1D 0 -profile_cache 1 -load_profile 0
mkdir -p ~/task2/profile_run_LPS && cd ~/task2/profile_run_LPS
/accel-sim/accel-sim-framework/gpu-simulator/bin/release/accel-sim.out \
    -trace /accel-sim/accel-sim-framework/util/tracer_nvbit/LPS_traces/kernelslist.g \
    -config ../ECE786_NCSU/gpgpusim.config -config ../SM86_RTX3070/trace.config
```

Step 2 — Rebuild: After modifying the source files, rebuild the simulator from the `accel-sim-framework` directory:

```
make -C /accel-sim/accel-sim-framework/gpu-simulator
```

Step 3 — Bypass Run: Copy the 15 profile files to a new directory, update the config, and run from that directory so the simulator can find the profile files in the current working directory:

```
-gpgpu_gmem_skip_L1D 0 -profile_cache 0 -load_profile 1
mkdir -p ~/task2/bypass_run_LPS
cp ~/task2/profile_run_LPS/profile_sm*.txt ~/task2/bypass_run_LPS/
cd ~/task2/bypass_run_LPS && <run accel-sim.out with same args>
```

Repeat Steps 1 and 3 for BFS and LUD benchmarks by substituting the appropriate trace path (BFS_traces or LUD_traces). The source environment must be set up with: `source gpu-simulator/setup_environment.sh` inside the Docker container before running any simulation.

2.3 Results

Table 2 shows IPC results for all cache-unfriendly benchmark kernels across three configurations: baseline (no bypass), full bypass (gmem_skip_L1D), and profile-based bypass. Table 3 filters to show only the cache-unfriendly kernels for clarity. LPS achieves a +10.7% IPC improvement with profile-based bypass. BFS cache-unfriendly kernels (uid 5-8) show modest improvements. LUD lud_internal kernels show slight IPC degradation in early iterations because the global filter bypasses some blocks that would benefit from caching in the second run, as the thread block execution order differs between profiling and bypass runs — a known limitation of static profiling-based approaches noted in the ICS'15 paper.

Table 2: Profile-Based Bypass Results (All Cache-Unfriendly Benchmarks)

Benchmark	Kernel Name	UID	IPC	IPC w/ Bypass	IPC w/ Profile	Category
LPS	_Z13GPU_laplace3diiiiPfs_	1	371.5598	545.6426	411.3269	Unfriendly
BFS	_Z6KernelP4NodePiPbS2_S1_S2_i	5	25.3154	32.2584	25.4560	Unfriendly
BFS	_Z6KernelP4NodePiPbS2_S1_S2_i	6	17.6819	31.5100	17.8256	Unfriendly
BFS	_Z6KernelP4NodePiPbS2_S1_S2_i	7	19.1406	40.9166	19.1834	Unfriendly
BFS	_Z6KernelP4NodePiPbS2_S1_S2_i	8	34.4767	56.8447	36.8354	Unfriendly
LUD	_Z13lud_perimeterPfii	2	21.9486	22.8194	20.8860	Unfriendly
LUD	_Z12lud_internalPfii	3	279.1061	326.3870	270.9351	Unfriendly
LUD	_Z13lud_perimeterPfii	5	21.0558	22.0443	21.1332	Unfriendly
LUD	_Z12lud_internalPfii	6	261.4349	317.9300	261.1100	Unfriendly
LUD	_Z13lud_perimeterPfii	8	19.4577	20.5372	19.7507	Unfriendly
LUD	_Z12lud_internalPfii	9	229.8357	291.5305	231.1369	Unfriendly
LUD	_Z13lud_perimeterPfii	11	17.9376	18.9236	18.0102	Unfriendly
LUD	_Z12lud_internalPfii	12	209.5789	255.2592	201.0982	Unfriendly
LUD	_Z13lud_perimeterPfii	14	16.5699	17.4015	16.7342	Unfriendly
LUD	_Z12lud_internalPfii	15	193.1855	230.7529	194.9091	Unfriendly
LUD	_Z13lud_perimeterPfii	17	15.2230	15.8610	15.2270	Unfriendly
LUD	_Z12lud_internalPfii	18	181.3300	206.9248	188.1940	Unfriendly
LUD	_Z12lud_internalPfii	21	154.6429	179.6369	155.1091	Unfriendly
LUD	_Z13lud_perimeterPfii	23	12.2285	12.7309	12.3235	Unfriendly
LUD	_Z12lud_internalPfii	24	131.8909	152.3141	136.1608	Unfriendly
LUD	_Z12lud_internalPfii	27	114.0209	121.8041	114.2689	Unfriendly
LUD	_Z12lud_internalPfii	30	86.4081	93.7268	85.9032	Unfriendly
LUD	_Z12lud_internalPfii	33	65.2366	68.4110	65.3858	Unfriendly
LUD	_Z12lud_internalPfii	36	42.3833	44.4568	42.2203	Unfriendly